

0.1 Solution

Setup. Write $b_i = P/a_i = \prod_{j \neq i} a_j$, so we are studying $g = \gcd_i(b_i + 1)$. By definition $b_i \equiv -1 \pmod{g}$ for every i .

Step 1: All the a_i lie in one residue class mod g .

Since $b_i \equiv -1 \pmod{g}$, we have $\gcd(b_i, g) = 1$, so each b_i is a unit mod g . As every a_j appears as a factor of some b_i (with $j \neq i$, possible since $k \geq 2$), each a_j is a unit mod g .

For all i we have $a_i b_i = P$, hence $a_i b_i \equiv a_\ell b_\ell \pmod{g}$. Using $b_i \equiv b_\ell \equiv -1$ this gives $a_i \equiv a_\ell \pmod{g}$. So there is a common unit c with

$$a_i \equiv c \pmod{g} \quad \text{for all } i.$$

Then $-1 \equiv b_i = \prod_{j \neq i} a_j \equiv c^{k-1} \pmod{g}$, i.e. $c^{k-1} \equiv -1 \pmod{g}$.

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Step 2: $k \geq 4$ cannot beat $k = 3$.

The k numbers are distinct positive integers in a fixed class mod g , so their sum is at least $\sum_{j=0}^{k-1} (r_0 + jg) \geq g \cdot \frac{k(k-1)}{2}$ where $r_0 \geq 1$. For $k \geq 4$ this is $\geq 6g$, so

$$6g \leq \sum a_i < 3 \cdot 10^6 \implies g < 500000.$$

Thus any g above 500000 must come from $k = 3$.

Step 3: The constraint for $k = 3$.

Here (*) becomes $c^2 \equiv -1 \pmod{g}$. Let $r \in \{1, \dots, g-1\}$ be the residue of c ; then

$$r^2 \equiv -1 \pmod{g}, \quad \text{so } r^2 + 1 = mg \text{ for some integer } m \geq 1.$$

The three distinct positive integers $\equiv c \pmod{g}$ have sum at least $r + (r+g) + (r+2g) = 3(r+g)$, so $3(r+g) < 3 \cdot 10^6 \implies r + g \leq 999999$.

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Step 4: Upper bound $g \leq 998285$.

Suppose $g \geq 998286$. By (†), $r \leq 999999 - g \leq 1713$, hence $r^2 + 1 \leq 1713^2 + 1 = 2934370$. Since $r^2 + 1 = mg \geq m \cdot 998286$:

- If $m \geq 3$: $r^2 + 1 \geq 3 \cdot 998286 = 2994858 > 2934370$ — impossible.
- If $m = 1$: $g = r^2 + 1 \geq 998286 \implies r \geq 1000$ (as $999^2 = 998001$), so $r + g = r + r^2 + 1 \geq 1001001 > 999999$, contradicting (†).
- If $m = 2$: $g = \frac{r^2+1}{2} \geq 998286 \implies r^2 \geq 1996571 \implies r \geq 1414$ (as $1413^2 = 1996569$); r is odd so $r \geq 1415$, giving $r + g \geq 1415 + \frac{1415^2+1}{2} = 1415 + 1001113 = 1002528 > 999999$, again contradicting (†).

Every case is impossible, so $g \leq 998285$.

Step 5: Construction achieving $g = 998285$.

Take $r = 1413$, $g = \frac{1413^2 + 1}{2} = \frac{1996570}{2} = 998285$, and

$$a_1 = 1413, \quad a_2 = 1413 + g = 999698, \quad a_3 = 1413 + 2g = 1997983.$$

These are distinct, and their sum is $2999094 < 3 \cdot 10^6$. Each $a_i \equiv 1413 \pmod{g}$, and

$$1413^2 = 1996569 = 2g - 1 \equiv -1 \pmod{g}.$$

Therefore every b_i (a product of two of the a_j) satisfies $b_i \equiv 1413^2 \equiv -1 \pmod{g}$, so $g \mid b_i + 1$ for all i . Hence $\gcd_i(b_i + 1) \geq 998285$, and by Step 4 it equals 998285.

Conclusion. Steps 2–4 show no configuration gives $g > 998285$, and Step 5 attains it.

$g_{\max} = 998285$
